

ADITYA DNYANESHWAR DAHALE

Systems & Infrastructure Software Engineer | Rust, C++, Go, Linux Internals, Distributed Systems

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VERIFIED MERGED OPEN-SOURCE CONTRIBUTIONS (ACCEPTED & SHIPPED)

- **Tokio Async Runtime (tokio-rs/tokio)** | [PR #8405](#) — **Semaphore Integer Overflow & Permit Leak** ✓ MERGED
Resolved critical permit leak in SemaphorePermit::merge where counts exceeding u32::MAX wrapped and leaked permits on drop; converted storage to checked usize.
- **FoxIO JA4 Fingerprinting (FoxIO-LLC/ja4)** | [PR #312](#) & [#314](#) — **Zero-Allocation Header Filter & Hex Digest** ✓ 2x MERGED
Eliminated heap string allocations in HTTP/1 header checks (#312); optimized hash12 by directly hex-encoding 6-byte raw digest slices, cutting 2 allocations and avoiding encoding 26 unused bytes (#314).
- **smoltcp TCP/IP Stack (smoltcp-rs/smoltcp)** | [PR #1194](#) — **Zero-Allocation Wire Parser** ✓ MERGED
Engineered single-pass zero-allocation TcpOptionsSummary wire parser for standalone network stack, removing heap allocations on the packet wire-path to boost packet ingestion.
- **SmartCore ML & MaxMindDB (smartcore, maxminddb-rust)** | [PR #386](#) & [#120](#) — **Memory & Instruction Optimizations** ✓ 2x MERGED
Pre-allocated exact vector capacities in Random Forest trees (#386); accelerated GeolIP lookups with native big-endian byte decoding (u32::from_be_bytes) for single-instruction node decoding (#120).

PARADIGM CORE ETHEREUM INFRASTRUCTURE (RETH • ALLOY • FOUNDRY)

- **Paradigm Reth (paradigmxyz/reth)** | [PR #27049](#) — **Snapshot Manifest Validation & Zero-Division Bounds** Open • Review Requested
Hardened SnapshotManifest::validate() to reject malformed manifests with blocks_per_file == 0 prior to expansion. Guarded ChunkedArchive::num_chunks() with saturating math to prevent snapshot panics.
- **Alloy Core SDK (alloy-rs/alloy)** | [PR #4191](#) — **Pending Transaction Heartbeat Recovery** Open • Review Requested
Fixed silent stream freeze in PendingTransactionBuilder::watch when pollers skip mined blocks during network jitter. Scans skipped block ranges across heartbeats to guarantee pending transactions complete.
- **Foundry (foundry-rs/foundry)** | [PR #16707](#) — **Chisel Session Isolation & Path Traversal Guard** Open • Review Requested
Hardened session validation in crates/chisel against Windows NTFS Alternate Data Streams (chisel-foo:bar.json) and path traversals, preventing arbitrary disk write escapes.

TIER-1 SYSTEMS RUNTIME & CLUSTER INFRASTRUCTURE (HYPER • FLY.IO • PINGORA)

- **Hyper HTTP Engine (hyperium/hyper)** | [PR #4183](#) — **RFC 6585 max_header_size Protocol Enforcement** Open • In Review
Added server/client builder APIs, strict RFC 6585 §5 (431 Too Large) rejection, stream buffer zero-copy guards, and chunked trailer decoder limits across 180+ tests.
- **Fly.io Cluster Infra (superfly/corrosion, ltx, macaroon)** | [PRs #578](#), [#102](#), [#104](#), [#48](#) — **Distributed Engine Resilience** In Review / CI Passed
Fixed schema migration crash panics in Corrosion (#578); fixed zero-page checksums in LTX (#104); enforced stream decoder bounds (#102); eliminated flaky deadline backoff bounds in Macaroon auth (#48).
- **Cloudflare Pingora & Turso libSQL** | [PR #996](#) & [PR #2280](#) — **Reverse Proxy & WAL Concurrency** CI 100% Green
Mapped upstream connect/TLS timeouts to 504 Gateway Timeout in Cloudflare Pingora; enforced default WAL journal concurrency for local libSQL instances.

FEATURED SYSTEMS ARCHITECTURE: SPRYZEN+ (4.83M RPS WIRE ENGINE)

- **Spryzen+ — Zero-Allocation Network Security & Telemetry Engine** | *Rust, Lock-Free Ring Buffers, SIMD, Tokio, Rayon Bulkheads, Docker*
 - **4.83M RPS & 207 ns P99 Latency:** Zero dynamic heap allocations on wire path via borrowed stack slices (Cow::Borrowed).
 - **27.99 ns Vectorized Normalization:** 256-byte compile-time LUT (NEEDS_DEEP_NORM_TABLE) with single-instruction byte lookups.
 - **Lock-Free Concurrency & Rayon Bulkheads:** Atomic ring queues eliminating thread locks; isolated jumbo payloads (>8KB) to Rayon workers.

AGENTIC OPERATIONS, INFRASTRUCTURE & AUTOMATION

- **AI-Native Engineering & Agent Tooling:** Built custom MCP (Model Context Protocol) servers and deterministic agent orchestration tools enabling coding agents to safely investigate test flakes, execute build verification, and automate release pipelines.
- **Observability & Telemetry:** Deployed Prometheus, Grafana, and Loki telemetry stacks with automated alerting guardrails; codified operational runbooks into programmatic verification harnesses to minimize incident blast radius.

TECHNICAL TOOLBELT & SYSTEMS PROFICIENCIES

- **Languages:** Rust (Advanced / Unsafe / Systems), C/C++ (C++17/20), Go, Python, Bash, SQL, Assembly basics
- **Systems & Architecture:** Zero-Allocation Wire Parsing, Lock-Free Ring Buffers, SIMD / LUTs, Rayon Bulkheads, eBPF/XDP, Tokio Runtime
- **Infrastructure & DevOps:** Linux Kernel & systemd, Docker, Kubernetes, CI/CD Workflows, Prometheus, Grafana, Loki, Terraform, MCP Servers
- **Networking & Protocols:** TCP/IP Wire Stacks (smoltcp), HTTP/1 & HTTP/2 (RFC 9110/6585), TLS 1.3 / JA4 Fingerprinting, Reverse Proxies

EDUCATION

Savitribai Phule Pune University (SPPU) — Bachelor of Engineering (B.E.) in Computer Engineering

Pune, India

Cumulative SGPA: 8.32 / 10.0 | Core Focus: Systems Programming, Operating Systems, Computer Architecture, Data Structures